Why i exists, and what it forces

The quadratic $x^2+x+1=0$ has discriminant -3: the reals are not enough. Adjoin one symbol i with the single property $i^2=-1$; a **complex number** is $z=a+bi$ with $a,b\in\mathbb{R}$. Here $a=\mathrm{Re}\,z$ and $b=\mathrm{Im}\,z$ — the imaginary part is b, **not** bi.

SHORT DERIVATION

The product rule is forced, not chosen

Expand with distributivity and reduce $i^2=-1$:

$$(a+bi)(c+di)=ac+adi+bci+bdi^2$$

$$(ac-bd)+(ad+bc)i.$$

No convention to memorise: every rule of the subject is "expand, then replace i^2 by -1". Equality is componentwise: $a+bi=c+di\Leftrightarrow a=c$ and $b=d$; a complex number is zero iff both parts are.

KEY RESULT

The conjugate, the division trick, and why $\mathbb{C}$ is a field

$\bar z=a-bi$. Then $z+\bar z=2\mathrm{Re}\,z$, $z-\bar z=2i\mathrm{Im}\,z$, $z\bar z=a^2+b^2=|z|^2\geq 0$, $=0$ iff $z=0$.

Division: for $w\neq 0$, multiply by $\bar w/\bar w$: $\frac{z}{w}=\frac{z\bar w}{|w|^2}$; in particular $\frac{1}{a+bi}=\frac{a-bi}{a^2+b^2}$. The conjugate exists to kill the imaginary denominator.

No zero divisors: if $w\neq 0$ and $zw=0$, multiply by $\bar w$: $z|w|^2=0\Rightarrow z=0$. Hence $\mathbb{C}$ is a field — factoring, Vieta, the quadratic formula all run unchanged. (And every complex number has a square root — sheet 04 — so every quadratic splits.)

WORKED EXAMPLE

Worked arithmetic — where the sign flips

$(3+2i)(5-i)=15-3i+10i-2i^2=17+7$ — the step $-2i^2=+2$ is the single most common slip.

$\frac{1+2i}{3-i}=\frac{(1+2i)(3+i)}{(3-i)(3+i)}=\frac{3+i+6i+2i^2}{9+1}=\frac{1+7i}{10}$.

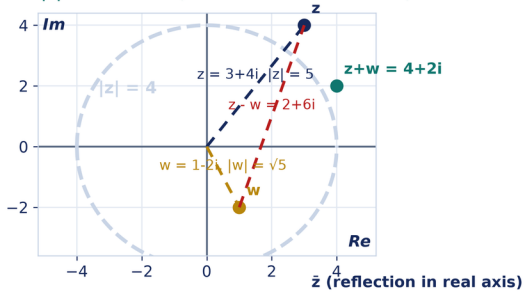
$\left(\frac{1+i}{1-i}\right)^3$: first $\frac{1+i}{1-i}=\frac{(1+i)^2}{(1-i)(1+i)}=\frac{2i}{2}=i$; then $i^3=-i$. **Reduce the base before the exponent.**

Cycle of i: $i^n=i^r$ with $r\in\{0,1,2,3\}$ the remainder of n on division by 4 — $(1+i)^{40}=((1+i)^2)^{20}=(2i)^{20}=2^{20}i^{20}=2^{20}$.

The plane: complex numbers are vectors

Plot $z=a+bi$ as (a,b). Addition is **vector addition**; the modulus $|z|=\sqrt{a^2+b^2}$ is the distance from the origin; $|z-w|$ is the distance between points; $\frac{z_1+z_2}{2}$ is the midpoint.

The Argand plane: |z| is a distance, z - w is the vector w → z, z̄ is the mirror



KEY RESULT

Triangle and reverse triangle inequalities — with equality conditions

$$|z+w|\leq|z|+|w|,\qquad ||z|-|w||\leq|z-w|$$

Geometric proof: the points $0, z, z+w$ form a (possibly degenerate) triangle; a side never exceeds the sum of the other two. **Equality** in the first iff z, w point the same way (one is a non-negative real multiple of the other) or one is 0; in the second, iff they point opposite ways. These equality conditions are the engine of every minimisation in sheet 10.

KEY RESULT

Parallelogram law (the hidden JEE workhorse)

$$|z+w|^2 + |z-w|^2 = 2|z|^2 + 2|w|^2$$

Proof: $|z \pm w|^2 = (z \pm w)(\bar{z} \pm \bar{w}) = |z|^2 + |w|^2 \pm (z\bar{w}$; add and the cross terms cancel. Geometry: sum of squared diagonals = sum of squared sides. Recognise it inside "minimise $|z|^2 + |z-2|^2$"-style targets and stop computing.

KEY RESULT

The standard Arg table (instant recall)

$$\operatorname{Arg} 1 = 0, \quad \operatorname{Arg} i = \tfrac{\pi}{2}, \quad \operatorname{Arg}(-1) = \pi,$$

$$\operatorname{Arg}(-i) = -\tfrac{\pi}{2}$$

$$\operatorname{Arg}(1+i) = \tfrac{\pi}{4}, \quad \operatorname{Arg}(-1+i) = \tfrac{3\pi}{4},$$

$$\operatorname{Arg}(1 - i\sqrt{3}) = -\tfrac{\pi}{3}$$

(Read $\operatorname{Arg}(-1+i) = \pi - \tfrac{\pi}{4} = \tfrac{3\pi}{4}$ — quadrant II, not $-\tfrac{\pi}{4}$.)

COMMON TRAP

Common traps — foundations

- $-2i^2 = +2$. Re-derive any forgotten rule by expanding and reducing $i^2 = -1$.
- $\operatorname{Im}(3+4i) = 4$, not $4i$.
- $\frac{z_1}{z_2} = 1$ means $z_1 = z_2$; $|\frac{z_1}{z_2}| = 1$ means only equal moduli.
- $|z-a| = r$ has center at a — the minus points **to** the center.
- $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$, $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$, $\bar{\bar{z}} = z$ — you may conjugate whole equations (used in sheet 07).
- $|z|^2 = z\bar{z} \neq z^2$ unless z is real.

ASSUMPTIONS & CONVENTIONS

Conventions for arg and Arg

- $\arg z$ (lowercase) = the full set of angles; statements like $\arg(z_1 z_2) = \arg z_1 + \arg z_2$ mean **equality of sets mod** 2π.
- Arg (capitalised) = the principal value in $(-\pi, \pi]$; it is not additive: $\operatorname{Arg}(z_1 z_2) = \operatorname{Arg} z_1 + \operatorname{Arg} z_2 - 2\pi\varepsilon$ with $\varepsilon \in \{-1, 0, 1\}$ chosen to land in the window.
- $z = 0$ has no argument; $\arg \bar{z} = -\arg z$, $\arg(1/z) = -\arg z$, $\arg(z^n) = n \arg z$ (as sets).

The argument: the direction half of the data

For $z = a + bi \neq 0$, an **argument** is any θ with $a = |z|\cos\theta$, $b = |z|\sin\theta$; arguments form the set $\arg z = \{\theta_0 + 2k\pi : k \in \mathbb{Z}\}$ (multi-valued — angles add only mod 2π). The **principal argument** is the canonical choice

$$\operatorname{Arg} z \in (-\pi, \pi].$$

Note the half-open window: $\operatorname{Arg}(-1) = \pi$, never $-\pi$; Arg of a fourth-quadrant point is negative. The quadrant of (a, b) fixes the reference angle — that check is the whole skill.

Polar form

Every $z \neq 0$ is uniquely $z = r(\cos\theta + i\sin\theta)$ with $r = |z| > 0$, $\theta \in \arg z$; the principal polar form takes $\theta = \operatorname{Arg} z$. Round trip check: expand $r\cos\theta + ir\sin\theta$ to get $a + bi$ back.

SHORT DERIVATION

Why products add angles — the calculation that seeds the subject

Let $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$, $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$. Expanding:

$$z_1 z_2$$

$$z_1 z_2 = r_1 r_2[(\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + i(\cos\theta_1\sin\theta_2 + \sin\theta_1\cos\theta_2)]$$

$$= r_1 r_2(\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2))$$

by the addition formulas from trigonometry. **Multiplication stretches by $r_1 r_2$ and rotates by $\theta_1 + \theta_2$.** Division: divide moduli, subtract arguments. This is "multiplication is rotation" made exact, and it is where $\arg(z_1 z_2) = \arg z_1 + \arg z_2$ comes from.

WORKED EXAMPLE

Worked — convert $-\sqrt{3}+i$ and $1-i\sqrt{3}$, with checks

$-\sqrt{3}+i$: $r = 2$, quadrant II, reference $\tan^{-1}\frac{1}{\sqrt{3}} = \frac{\pi}{6}$, so $\theta = \frac{5\pi}{6}$:

$$-\sqrt{3}+i = 2\left(\cos\tfrac{5\pi}{6} + i\sin\tfrac{5\pi}{6}\right).$$

Back: $2\cos\frac{5\pi}{6} = -\sqrt{3}$ ✓, $2\sin\frac{5\pi}{6} = 1$ ✓.

$1-i\sqrt{3}$: $r = 2$, quadrant IV, $\text{Arg} = -\frac{\pi}{3}$ — the same magnitude as $\frac{\pi}{3}$ but **negative**: $1-i\sqrt{3} = 2(\cos(-\frac{\pi}{3}) + i\sin(-\frac{\pi}{3}))$.

$\frac{1}{1+i} = \frac{1-i}{2} = \frac{1}{\sqrt{2}}(\cos(-\frac{\pi}{4}) + i\sin(-\frac{\pi}{4}))$ — check via the rules: $|1/(1+i)| = 1/\sqrt{2}$ ✓, $\arg = -\arg(1+i) = -\frac{\pi}{4}$ ✓. Both languages agree; use one to check the other.

KEY RESULT

Multiplication by i and by $e^{i\theta}$ — the rotation reflex

$i(a+bi) = -b+ai$: the point $(a,b) \mapsto (-b,a)$ — length preserved (a^2+b^2 unchanged; $|iz| = |i||z| = |z|$), dot product $(a,b)\cdot(-b,a) = 0$, orientation $(1,0) \mapsto (0,1)$: a 90° counterclockwise turn. So $\times i$ = quarter-turn CCW, $\times(-1)$ = half-turn, $\times(-i)$ = quarter-turn clockwise, $\times re^{i\theta}$ = stretch r + rotate θ.

STANDARD APPLICATION

Where this pays off — the two-language method

Translate the problem into whichever language makes it visible, solve, translate back. "Solve $iz = \bar{z}$": algebra $z = a+bi$, $iz = -b+ai = a-bi = \bar{z} \Rightarrow a = -b$: the line $y = -x$. Geometry: rotating z by 90° gives its mirror — the line at -45°. Same answer, two sentences; if they disagree, one of them is wrong — that is the check.

COMMON TRAP

Common traps — argument and polar

- Principal window is $(-\pi, \pi]$: $\operatorname{Arg}(-1) = \pi$; $\operatorname{Arg}(-i) = -\frac{\pi}{2}$; fourth-quadrant arguments are negative.
- "$\arg z = \theta$" statements are mod 2π; when a problem asks for the argument it usually means Arg — state which you are giving.
- Arg is not additive across products; $\arg$ (the set) is. Using principal values in De Moivre without reducing mod 2π is the classic wrong-answer machine.
- $z = r(\cos\theta + i\sin\theta)$ with $r > 0$; writing $r < 0$ just shifts θ by π — keep r positive and θ absorbing the sign.
- $\operatorname{Arg}(z_1/z_2) \neq \operatorname{Arg} z_1 - \operatorname{Arg} z_2$ blindly: $z_1 = 1, z_2 = -1$ gives $\operatorname{Arg}(-1) = \pi$ while the naive difference $0 - \pi = -\pi$ sits outside the window. The correct statement: $\operatorname{Arg}(z_1/z_2) \equiv \operatorname{Arg} z_1 - \operatorname{Arg} z_2 \pmod{2\pi}$, then **reduce into** $(-\pi, \pi]$.

De Moivre's theorem

KEY RESULT

Statement (for integer $n \geq 1$, extended to all $n \in \mathbb{Z}$)

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$$

$$(r(\cos\theta + i\sin\theta))^n = r^n(\cos n\theta + i\sin n\theta).$$

SHORT DERIVATION

Proof — one induction step

$n = 1$ is trivial. If it holds for n, multiply by $(\cos\theta + i\sin\theta)$ and use the angle-addition calculation of sheet 02: $(\cos n\theta + i\sin n\theta)(\cos\theta + i\sin\theta) = \cos(n+1$. ∎

Negative n: $(\cos\theta + i\sin\theta)^{-1} = \cos\theta - i\sin\theta = \cos(-\theta)$ (conjugate flips the angle), so $z^{-m} = (z^{-1})^m$ extends the theorem to all integers. **Integer exponents only** — rational powers are multi-valued and go through the n-th-root machinery of sheet 04, never through "De Moivre with fractions".

WORKED EXAMPLE

Worked — $(1+i)^{10}$ and $(\sqrt{3}+i)^4$

$1 + i = \sqrt{2}\, e^{i\pi/4}$ (Euler form, below), so $(1+i)^{10} = 2^5 e^{i10\pi/4} = 32e^{i5\pi/2} = 32e^{i\pi/2} = 32$ — the only real work is reducing $5\pi/2 \equiv \frac{\pi}{2} \pmod{2\pi}$.

$(\sqrt{3}+i)^4$: $r = 2$, $\theta = \frac{\pi}{6}$: $2^4(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}) = 16(-\frac{1}{2} + i\frac{\sqrt{3}}{2}) = -8 + 8$. Brute-force check by squaring twice gives the same — De Moivre halves the work at each step.

Pattern bank: $(1+i)^2 = 2i$, $(1+i)^4 = -4$, $(1+i)^8 = 16$ — the direction cycles with period 8 ($n \cdot 45^\circ$).

SHORT DERIVATION

Triple angles in ten seconds — the trig payoff

Set $n = 3$ and expand $(\cos\theta + i\sin\theta)^3 = \cos^3\theta + 3i\cos^2\theta\sin\theta - 3$. Equate real and imaginary parts (componentwise equality):

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta,$$

$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta.$$

Every "hard" multiple-angle identity is a De Moivre expansion you can redo in seconds — do not memorise these; derive them.

Euler form

Define $e^{i\theta} := \cos\theta + i\sin\theta$ (for real θ). Then $z = re^{i\theta}$, $z^n = r^n e^{in\theta}$, roots are $R^{1/n}e^{i(\phi+2k\pi)/n}$, and the notation is exact (expanding the exponential series at $i\theta$ groups into $\cos\theta + i\sin\theta$ term by term). The famous checkpoint: $e^{i\pi} + 1 = 0$.

KEY RESULT

The Euler-form move table

Powers: $(re^{i\theta})^n = r^n e^{in\theta}$ (multiply the angle).
Products: multiply moduli, add angles.
Quotients: divide moduli, subtract angles.

The workhorse pair (for $z = re^{i\theta}$, and for $|z| = 1$ where $\bar{z} = 1/z$):

$$z + \bar{z} = 2r\cos\theta, \qquad z - \bar{z} = 2ir\sin\theta,$$

$$e^{i\theta} + e^{-i\theta} = 2\cos\theta$$

$$e^{i\theta} - e^{-i\theta} = 2i\sin\theta, \qquad \cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2},$$

$$\sin\theta = \frac{e^{i\theta}-e^{-i\theta}}{2i}$$

STANDARD APPLICATION

Why $z^n + z^{-n}$ is a polynomial in $2\cos\theta$

For $z = e^{i\theta}$: $z^n + z^{-n} = 2\cos n\theta$ — real, always. E.g. $z^3 + z^{-3} = 2\cos 3\theta = 8\cos^3\theta - 6\cos\theta$. This bridge (complex ↔ trig) is how olympiad trig-product identities and JEE "express $\cos 3\theta$ in terms of $\cos\theta$" questions are the same question. Also: three unit vectors 120° apart sum to zero — $e^{i\theta}(1 + \omega + \omega^2) = 0$ — the geometric seed of sheet 05.

ASSUMPTIONS & CONVENTIONS

Scope note on $e^{i\theta}$

$e^{i\theta}$ is defined as $\cos\theta + i\sin\theta$ here; the full complex exponential e^z for arbitrary complex z (series, log) is a calculus-topic treated later in the series. Everything on this sheet uses only the definition plus the algebra of sheets 01–02. The notation is used because it compresses De Moivre to one line; the content never depends on it.

COMMON TRAP

Common traps — De Moivre and Euler

- **Integer exponents only:** $(\cos\theta + i\sin\theta)^{1/2} \neq \cos\frac{\theta}{2} + i\sin\frac{\theta}{2}$ as a statement about *all* square roots — the right side is one of the two roots (the principal one), the other is its negative. Rational powers go through sheet 04.
- Reducing the angle: $(1+i)^{10} = 32i$ requires $10\pi/4 \equiv \pi/2 \pmod{2\pi}$; forgetting the reduction gives phantom directions.
- $e^{i\theta} = 1 \Leftrightarrow \theta \equiv 0 \pmod{2\pi}$; $e^{i\theta}$ has modulus 1 for real θ, so $|e^{i\theta}| = 1$ never gives magnitude information.
- Equating real/imaginary parts to extract $\cos n\theta$ needs the powers of $\sin^2 = 1 - \cos^2$ converted consistently — a messy sign here is not a "deep" error, it is bookkeeping.
- $\bar{z} = 1/z$ is valid **only** when $|z| = 1$; $z + 1/z = 2\cos\theta$ likewise.

Solving $z^n = w$

Write $w = R(\cos\phi + i\sin\phi)$ $(R > 0)$ and $z = r(\cos\theta + i\sin\theta)$ $(r > 0)$. De Moivre splits the equation into two real conditions: $r^n = R$ and $n\theta \equiv \phi \pmod{2\pi}$.

KEY RESULT

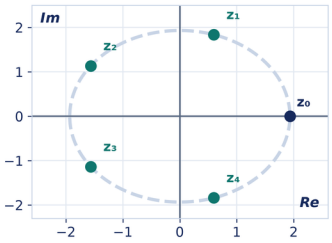
The n-th roots of $w \neq 0$ — count and geometry

$$z_k = R^{1/n}\left(\cos\frac{\phi + 2k\pi}{n} + i\sin\frac{\phi + 2k\pi}{n}\right),$$

$$k = 0, 1, \ldots, n-1.$$

Exactly n distinct roots: $r = R^{1/n}$ is unique, and $\theta = \frac{\phi + 2k\pi}{n}$ repeats mod 2π with period exactly n in k. They lie on the circle $|z| = R^{1/n}$ at equal angular steps $\frac{2\pi}{n}$: **the vertices of a regular n-gon**. Consecutive roots differ by the factor $e^{2\pi i/n}$.

The five 5th roots of 27 on |z| = 27^{1/5} ≈ 1.933, steps of 72° — a regular pentagon



WORKED EXAMPLE

Worked — cube roots of 8 and fourth roots of 16

Cube roots of 8: $R = 8, \phi = 0$; $z_k = 2(\cos\frac{2k\pi}{3} + i\sin\frac{2k\pi}{3})$, $k = 0, 1, 2$:

$$z_0 = 2, \qquad z_1 = -1 + i\sqrt{3}, \qquad z_2 = -1 - i\sqrt{3}.$$

Check z_1^3: $-1 + i\sqrt{3} = 2e^{2\pi i/3}$, cube $= 8e^{2\pi i} = 8$ ✓.

Fourth roots of 16: $r = 2$, angles $\frac{k\pi}{2}$: $\{2, 2i, -2, -2i\}$ — the square inscribed in $|z| = 2$.

Cube roots of -27: $R = 27$, $\phi = \pi$: $\{-3, \frac{3}{2} \pm i\frac{3\sqrt{3}}{2}\}$ — equilateral triangle centered at 0; Vieta on $t^3 + 27 = 0$ immediately gives $z_1 + z_2 + z_3 = 0$ and $z_1z_2 + z_2z_3 + z_3z_1 = 0$ — the geometry explains Vieta here.

Square roots of an arbitrary complex number — the explicit formula

KEY RESULT

Formula (derived, not guessed)

For $w = a + bi$, $r = |w| = \sqrt{a^2 + b^2}$: the two square roots of w are

$$\pm\left(\sqrt{\frac{r+a}{2}} + i\,\mathrm{sgn}(b)\sqrt{\frac{r-a}{2}}\right) \qquad (b \neq 0)$$

with the sign of the imaginary part tied to the sign of b (for $b = 0$: $\sqrt{a}$ or $i\sqrt{-a}$ by the sign of a).

SHORT DERIVATION

Derivation — match parts

Seek $\sqrt{a+bi} = x+iy$ with $x, y \in \mathbb{R}$. Squaring: $(x+iy)^2 = x^2 - y^2 + 2xyi = a+bi$, so $x^2 - y^2 = a$, $2xy = b$. Also $|x+iy|^2 = x^2+y^2 = |w| = r$. Add and subtract:

$$x^2 = \frac{r+a}{2}, \qquad y^2 = \frac{r-a}{2}$$

(both ≥ 0 since $|a| \leq r$), and $xy = \frac{b}{2}$ forces x, y to share the sign of b (with $x > 0$ by convention). ∎

Worked: $\sqrt{3+4i}$: $r = 5$; $x^2 = 4$, $y^2 = 1$, $b > 0$: roots $\pm(2+i)$ ✓ ($(2+i)^2 = 3+4i$). $\sqrt{-3+4i}$: $r = 5$; $x^2 = 1$, $y^2 = 4$: roots $\pm(1+2i)$ ✓. $\sqrt{4i}$: $r = 4$, $a = 0$; $x^2 = y^2 = 2$, $b > 0$: $\pm(\sqrt{2}+i\sqrt{2})$ ✓ ($(\sqrt{2}+i\sqrt{2})^2 = 2(1+i)^2 = 4i$).

COMMON TRAP

Common traps — roots

- **"Find the cube roots"** means all n of them: "the cube root of 8" as a complete answer is $\{2, -1 \pm i\sqrt{3}\}$.
- $\sqrt{w}$ written bare usually means the **principal** root (arg in $(-\frac{\pi}{2}, \frac{\pi}{2}]$); "find the square roots" means both.
- **Never distribute roots over products for complex numbers**: $\sqrt{(-1)^2} = 1$ but $(\sqrt{-1})^2 = -1$; $\sqrt{z}\sqrt{w} \neq \sqrt{zw}$ in general. Solve $z^n = w$ from scratch instead.
- The sign in the square-root formula is $\operatorname{sgn}(b)$ on the imaginary part (and $x > 0$) — mixing it up gives a root of $\bar{w}$ instead of w.
- Roots of 0: only $z = 0$ (the formula and the count assume $w \neq 0$).
- Rational exponents $w^{p/q}$: reduce p/q to lowest terms, then read as "q-th roots raised to p" — exactly the parity discipline of chapter 2's rational exponents.

The cube roots of unity

$z^3 = 1$ has solutions $1, \omega = e^{2\pi i/3} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}, \omega^2 = e^{4\pi i/3} = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$. **By universal convention ω denotes a non-real cube root of unity**: $\omega^3 = 1$, $\omega \neq 1$.

KEY RESULT

The five ω-identities (each with its one-line proof)

1. $1 + \omega + \omega^2 = 0$ — since $\omega \neq 1$ is a root of $x^3 - 1 = (x-1)(x^2+x+1)$.
2. $\bar{\omega} = \omega^2$ — conjugate reflects in the real axis; ω, ω^2 sit at $\pm 120°$.
3. $(1-\omega)(1-\omega^2) = 3$ — expand: $1 - (\omega + \omega^2) + \omega^3 = 1 + 1 + 1 = 3$.
4. $1 + \omega = e^{i\pi/3}$ — compute: $\frac{1}{2} + i\frac{\sqrt{3}}{2}$; modulus 1, argument $60°$; also $1 + \omega = -\omega^2$.
5. $1 - \omega = \sqrt{3}\, e^{-i\pi/6}$ — so $|1-\omega| = \sqrt{3}$ (the side of the triangle) and $\arg(1-\omega) = -30°$.

Drill: $\omega + \omega^2 = -1$, $\omega\omega^2 = \omega^3 = 1$, $\omega^{3k} = 1$.

SHORT DERIVATION

Why the triangle geometry keeps reappearing

$1, \omega, \omega^2$ are equally spaced on the unit circle: an equilateral triangle. Identity 3 says the squared side length is 3; identity 4 says $1 + \omega$ is the vertex reached by rotating 1 by $60°$. Every ω-computation in an exam is either "expand and use $1 + \omega + \omega^2 = 0$, $\omega^3 = 1$" or "read the geometry". Try both when stuck — they must agree.

WORKED EXAMPLE

Worked — $(1+\omega)(1+\omega^2)$, and the sign trap

Algebra: $1+(\omega+\omega^2)+\omega^3 = 1-1+1 = 1$. Geometry: $(e^{i\pi/3})(e^{-i\pi/3}) = 1$ ✓. The classic wrong answer is 2 — produced by treating $\omega+\omega^2$ as $+1$. If you get 2, the sign drill above has slipped.

Also: $\sum_{k=0}^{2023}\omega^k$: $2024 = 3\cdot 674 + 2$ terms; each block of three sums to 0, leaving $\omega^{2022}+\omega^{2023} = 1+\omega = -\omega^2$.

$(\omega+\omega^2)^3 + 3(\omega+\omega^2) + 1 = (-1)^3 + 3(-1)$.

The n-th roots of unity and the vanishing sum

The n-th roots are $\zeta_k = e^{2\pi ik/n}$, $k = 0, \dots, n-1$ — a regular n-gon on $|z|=1$; write $\zeta = e^{2\pi i/n}$.

KEY RESULT

Vanishing-sum identity — the engine of every sum-of-roots problem

For integers m, n with $n \geq 1$:

$$\sum_{k=0}^{n-1}\zeta^{mk} = n \text{ if } n \text{ divides } m, \qquad = 0 \text{ otherwise.}$$

Proof: geometric series with ratio ζ^m: if $n \nmid m$ then $\zeta^m \neq 1$ and the sum is $\frac{(\zeta^m)^n - 1}{\zeta^m - 1} = \frac{(\zeta^n)^m - 1}{\zeta^m - 1} = 0$; if $n \mid m$ every term is 1. ∎

Read geometrically: the vertices of a regular n-gon centered at 0 sum to zero. Corollaries: $\sum_{k=0}^{n-1}\zeta^k = 0$ for $n > 1$; the sixth-roots sum $\sum_{k=0}^{5} e^{i\pi k/3} = 0$; a geometric progression of unit steps around the circle always closes.

STANDARD APPLICATION

Standard applications — the two reflexes

1. **Sums of powers:** $\sum_{k=0}^{N}\omega^k$-type: reduce the number of terms mod 3 (or mod n), leftover terms only.
2. **Vieta on $x^n - 1$:** the coefficient of x^{n-1} is 0, so $\zeta_0 + \cdots + \zeta_{n-1} = 0$ — the vanishing sum again. For $t^3 + 27 = 0$ (cube roots of -27): sum $= 0$, pairwise sum $= 0$ — see sheet 04. Whenever roots sit on a regular polygon, Vieta's zero coefficients are the polygon's symmetry in disguise.

ASSUMPTIONS & CONVENTIONS

Conventions

ω = a **non-real** cube root of unity ($\omega = e^{2\pi i/3}$ or $e^{4\pi i/3}$ — the two are ω and ω^2, and "ω is a non-real cube root" statements must hold for either choice). $\zeta = e^{2\pi i/n}$ for general n. "The n-th roots of unity" always means the full set of n. All sums in this sheet are finite; no convergence questions arise.

COMMON TRAP

Common traps — roots of unity

- $(1+\omega)(1+\omega^2)=1$, not 2. ($\omega+\omega^2=-1$.)
- $\omega^3=1$ does **not** let you "cancel cubes" inside sums like $\omega^2+\omega+1$ — that sum is 0 by identity I, not by cancellation.
- $\sum_{k=0}^{n-1}\zeta^{mk}=0$ needs $n\nmid m$; e.g. $\sum_{k=0}^{5}e^{i\pi k/3}$ is 0 because the terms are the six sixth-roots of unity — the exponent step is $\pi/3=2\pi/6$ ✓ — but $\sum_{k=0}^{5}e^{i\pi k/6}$ is **not** 0 (that is twelve steps' worth compressed into six terms: a geometric series with ratio $e^{i\pi/6}$, sum $\frac{1-e^{i\pi}}{1-e^{i\pi/6}}=\frac{2}{1-e^{i\pi/6}}\neq 0$).
- $\bar\omega=\omega^2$ is special to cube roots ($\omega^2=\omega^{-1}=\bar\omega$ needs $|\omega|=1$ ✓ but the exponent identity is $n=3$ specific).
- Using ω as if it were real: $\omega^2\neq\omega$; $\omega^2=\bar\omega$ is complex, and $|\omega|=1$ means $\omega\bar\omega=1$.

Products of $(1-\zeta_k)$ — where the number n hides

KEY RESULT

Product formula

For $\zeta_k=e^{2\pi ik/n}$ $(k=0,\dots,n-1)$:

$$\prod_{k=1}^{n-1}(1-\zeta_k)=n.$$

SHORT DERIVATION

Proof — factor, then let $x\to 1$

$$x^n-1=\prod_{k=0}^{n-1}(x-\zeta_k)=(x-1)\prod_{k=1}^{n-1}(x-\zeta_k).$$

Divide by $x-1$ and take $x\to 1$ (equivalently, differentiate at $x=1$):

$$\prod_{k=1}^{n-1}(1-\zeta_k)=\lim_{x\to 1}\frac{x^n-1}{x-1}=n.\ \blacksquare$$

Two checks: $n=3$: $(1-\omega)(1-\omega^2)=3$ ✓ (sheet 05, identity 3). $n=5$: $\prod_{k=1}^{4}(1-e^{2\pi ik/5})=5$ ✓. The factors come in conjugate pairs, so the product is real and positive — consistent.

KEY RESULT

Trigonometric cousin — the sine product

$$\prod_{k=1}^{n-1}\sin\frac{\pi k}{n}=\frac{n}{2^{n-1}}$$

Derivation: $|1-\zeta_k|=2\sin\frac{\pi k}{n}$ (compute: $|1-e^{2\pi ik/n}|=|e^{\pi ik/n}||e^{-\pi ik/n}-e^{\pi ik/n}|=2\sin\frac{\pi k}{n}$, positive for $1\le k\le n-1$). Take moduli in the product formula. Check $n=6$: $\frac12\cdot\frac{\sqrt3}{2}\cdot 1\cdot\frac{\sqrt3}{2}\cdot\frac12=\frac{3}{16}$ and $\frac{6}{2^5}=\frac{3}{16}$ ✓. E.g. $\sin\frac{\pi}{5}\sin\frac{2\pi}{5}\sin\frac{3\pi}{5}\sin\frac{4\pi}{5}=\frac{5}{16}$.

The roots-of-unity filter

Problem type: compute $\sum_{k\equiv r\pmod n}\binom{N}{k}$ — add only the terms whose index sits in one residue class.

KEY RESULT

The filter (derived on the spot if forgotten)

For $\zeta = e^{2\pi i/n}$, the average $\frac{1}{n}\sum_{j=0}^{n-1}\zeta^{j(k-r)}$ equals 1 when $n \mid (k-r)$ and 0 otherwise (vanishing sum — sheet 05). Hence for $P(x)=\sum_k c_k x^k$:

$$\sum_{k\equiv r \pmod n} c_k = \frac{1}{n}\sum_{j=0}^{n-1}\zeta^{-rj}P(\zeta^j)$$

and for $P(x)=(1+x)^N$:

$$\sum_{k\equiv r \pmod n}\binom{N}{k} = \frac{1}{n}\sum_{j=0}^{n-1} e^{-2\pi i rj/n}(1+e^{2\pi ij/n})^N.$$

WORKED EXAMPLE

Worked — $\sum_{k\equiv 0 \pmod 3}\binom{9}{k}$

Filter with $n=3$, $r=0$, $\omega = e^{2\pi i/3}$:

$$\tfrac{1}{3}[(1+1)^9+(1+\omega)^9+(1+\omega^2)^9]$$

$$= \tfrac{1}{3}[512+(e^{i\pi/3})^9+(e^{-i\pi/3})^9]$$

$$= \tfrac{1}{3}(512-1-1) = 170.$$

Direct check: $\binom{9}{0}+\binom{9}{3}+\binom{9}{6}+\binom{9}{9} = 1+84+84+1 = 17$ ✓.

Sibling classes: $r=1$: $\frac{1}{3}[512+\omega^{-1}(-1)+\omega^{-2}(-1)] = \frac{1}{3}(512+1) =$; $r=2$ likewise 171. Check: $170+171+171 = 512 = 2^9$ ✓ — the classes partition the power set.

STANDARD APPLICATION

When to reach for the filter

- Any "sum of binomial coefficients where $k \equiv r \pmod n$" — JEE Advanced has asked ω-flavoured versions repeatedly (e.g. sums of $\binom{N}{k}$ over $k\equiv 1 \pmod 3$).
- Evaluating $(1+\zeta^j)^N$: convert $1+\zeta^j$ to polar first ($1+\omega = e^{i\pi/3}$, $1+\omega^2 = e^{-i\pi/3}$, $1+i=\sqrt{2}e^{i\pi/4}$ …) then De Moivre.
- Alternating sums ($(-1)^k\binom{N}{k}$) are the case $n=2$: $\sum_{k\text{ even}}\binom{N}{k} = \sum_{k\text{ odd}}\binom{N}{k} = 2^{N-1}$ for $N\ge 1$.

SHORT DERIVATION

Side identity worth owning — $\prod_{k=0}^{n-1}|z-\zeta_k| = |z^n-1|$

Factor $z^n-1=\prod(z-\zeta_k)$ and take moduli: one line. At $z=2$, $n=5$: $\prod|2-\zeta_k| = |32-1| = 31$ — a five-distance product with no trigonometry. At $z=e^{i\theta}$ on the unit circle: $\prod_{k=0}^{n-1}|e^{i\theta}-\zeta_k| = |e^{in\theta}-1| = 2\left|\sin\frac{n\theta}{2}\right|$ — a clean chord product (olympiad sheets use this form).

COMMON TRAP

Common traps — products and filters

- $\prod_{k=1}^{n-1}(1-\zeta_k)=n$ is a **product of complex numbers**; its modulus interpretation ($|1-\zeta_k|=2\sin\frac{\pi k}{n}$) is a different statement — keep the two apart.
- Filter setup: the sum must run over $k\equiv r \pmod n$ **with the same n as the root of unity**; mixing $n=3$ filter with a mod-4 condition is the standard misfire.
- $(1+\omega)^9=-1$ requires $1+\omega=e^{i\pi/3}$ (identity 4) first — expanding $(1+\omega)^9$ by the binomial theorem is a trap door.
- In the sine product, the index runs 1 to $n-1$ (not n) and $\sin\frac{\pi k}{n}>0$ throughout — sign care when n is even.
- $\sum_{k\equiv 0\,(3)}\binom{9}{k}=170$ counts $k=0,3,6,9$ — $k=0$ is included; "positive multiples of 3" is a different (170 − 1) answer. Read the residue class exactly.

The exam's native equation type

Real-variable techniques assume the unknown appears "purely"; JEE mixes z and $\bar z$: $2z+i\bar z=5+2i$, or $|z|+z=1+i$. Two systematic methods exist; run the second as a check on the first.

KEY RESULT

Method 1 — real and imaginary parts (always works)

Write $z=a+bi$, $\bar z=a-bi$, expand, equate components. Two real linear equations in a,b — mechanical and bulletproof. Reach for it under pressure.

KEY RESULT

Method 2 — the conjugate trick (fast, linear equations only)

If the equation is linear in z and $\bar z$ — $\alpha z+\beta\bar z=c$ — conjugate it: $\bar\alpha\bar z+\bar\beta z=\bar c$. Now solve the 2×2 system in the unknowns $z,\bar z$. Watch the sign: $\bar i=-i$.

WORKED EXAMPLE

Worked — $2z+i\bar z=5+2i$ both ways

Method 1:
$2(a+bi)+i(a-bi)=(2a+b)+i(2b+a)=$, so $2a+b=5$, $a+2b=2$; subtract twice the second from the first: $-3b=1\Rightarrow b=-\frac{1}{3}$, $a=\frac{8}{3}$.

Method 2: conjugate: $2\bar z-iz=5-2i$.
Multiply by i: $2i\bar z+z=i(5-2i)=2+5i$.
Subtract from the original:
$(2z+i\bar z)-(z+2i\bar z)=z-i\bar z=3-3i$. Add to the original: $3z=8-i\Rightarrow z=\frac{8}{3}-\frac{1}{3}i$ ✓ same answer.

Check by substitution (mandatory):
$2(\frac{8}{3}-\frac{i}{3})+i(\frac{8}{3}+\frac{i}{3})=\frac{16}{3}-\frac{2i}{3}+\frac{8i}{3}-\frac{1}{3}=5+2i$ ✓.

STANDARD APPLICATION

The $z+\bar z=2\mathrm{Re}\,z$ family — linear in the parts

Equations like $z+\bar z=4$, $z-\bar z=6i$ solve by adding/subtracting: $2\mathrm{Re}\,z=4$, $2i\mathrm{Im}\,z=6i\Rightarrow z=2+3i$. "$2\bar z-3z$ is purely imaginary": expand $-a-5bi$; real part $=0\Leftrightarrow a=0$; the whole imaginary axis (including 0). "Purely imaginary $\Leftrightarrow z+\bar z=0$"; "real $\Leftrightarrow z-\bar z=0$"; "$|z|=r$ with $z=x+iy$": $x^2+y^2=r^2$.

Quadratics with real coefficients — conjugate roots

If $x^2+bx+c=0$ has real coefficients and discriminant $\Delta=b^2-4c<0$, the roots are $z_{1,2}=\frac{-b}{2}\pm i\frac{\sqrt{4c-b^2}}{2}$ — automatically conjugates. JEE asks for *quantities built from* the roots.

KEY RESULT

Root-distance formulas (no root-finding needed)

For $x^2+bx+c=0$, b, c real, $\Delta<0$: with $z_2=\bar{z}_1$:

$$|z_1-z_2|=\sqrt{4c-b^2}=\sqrt{|\Delta|},$$

$$|z_1-\bar{z}_1|=\sqrt{4c-b^2}\ \text{(same number)}$$

$$|z_1|=\sqrt{c}\ (\text{when } c>0), \qquad \operatorname{Re} z_1=-\tfrac{b}{2}$$

Worked: $x^2-4x+13=0$: $\Delta=-36$, roots $2\pm 3i$: $|z_1-z_2|=|6i|=6=\sqrt{36}$ ✓, $|z_1|=\sqrt{13}$ ✓. The point of the formulas: never compute the roots themselves when the question is a modulus/distance.

SHORT DERIVATION

Why conjugate roots — one line

Complex roots of a real polynomial come in conjugate pairs because conjugating $p(z)=0$ (legal: conjugation respects sums and products, $\bar{1}=1$) gives $p(\bar{z})=0$. So $\bar{z}_1$ is a root whenever z_1 is; for a quadratic the two roots are exactly $\{z_1, \bar{z}_1\}$ (distinct when $\Delta\neq 0$). The same argument locks every degree: real polynomials have conjugate-symmetric root sets — Vieta's real coefficients are the algebraic shadow of the reflection symmetry.

ASSUMPTIONS & CONVENTIONS

Scope and conventions

- "z and $\bar{z}$ equations" are solved over $\mathbb{C}$ with $z=a+bi$, $a, b\in\mathbb{R}$; "purely imaginary" includes 0 unless a problem excludes it explicitly.
- Quadratics in this sheet have **real** coefficients (stated each time); complex coefficients can have non-conjugate root pairs — then only Method 1 applies.
- $|z_1-z_2|=\sqrt{|\Delta|}$ assumes $|a|=1$ leading coefficient; for ax^2+bx+c scale first.

COMMON TRAP

Common traps — equations

- Method 2's conjugation step: $i\bar{z}=-iz$, not iz. A sign slip here poisons both methods' agreement.
- "Purely imaginary" allows 0; "imaginary" sometimes means "non-real" in problem text — read the convention in the question.
- $|z_1-z_2|=\sqrt{|\Delta|}$ needs $\Delta<0$ and real coefficients; with $\Delta>0$ the roots are real and $|z_1-z_2|=\sqrt{\Delta}$ (no conjugates involved).
- Cancelling z or $\bar{z}$ terms across an equation without checking $z=0$ cases: $z\bar{z}=|z|^2$ vanishes only at $z=0$ — test it.
- $|z|+z=1+i$-type: modulus equations are **non-linear** — Method 2 fails; split into parts and use $|z|=\sqrt{a^2+b^2}\geq 0$ as an extra unknown with a consistency condition ($a^2+b^2=$ the squared equation).

Reading loci off the page

$|z-z_0|=r$ **is** the circle of center z_0 and radius r; the exam skill is recognising circle conditions in disguise.

KEY RESULT

The locus dictionary (modulus side)

$|z-z_0|=r$ circle; $\quad |z-z_0|<r$ open disk

$|z-a|=|z-b|$ perpendicular bisector of ab;

$\operatorname{Re} z=c$ vertical line;

$\operatorname{Im} z=d$ horizontal line

$|z-a|=k|z-b|$ $(k\neq 1)$ circle (Apollonius);

$|z-a|+|z-b|=2a_0$ ellipse (if $2a_0>|a-b|$)

$|z-a|+|z-b|=|a-b|$ the segment ab itself;

$<|a-b|$ empty set

SHORT DERIVATION

Apollonius circle — derived once, used forever

$|z-a| = k|z-b|$, square:
$|z|^2 - z\bar{a} - \bar{z}a + |a|^2 = k^2(|z|^2 - z\bar{b} - \bar{z}b + |b$
. Collect (with $k \neq 1$):
$(1-k^2)|z|^2 - z(\bar{a} - k^2\bar{b}) - \bar{z}(a - k^2 b) + |a|^2$
— divide by $(1-k^2)$ and complete the square: a circle. **Special case** $k = 1$: **the circle degenerates to the perpendicular bisector** (the $|z|^2$ terms cancel). The limit case is not a formality; exam problems love $k = 1$ hidden as "equidistant from".

WORKED EXAMPLE

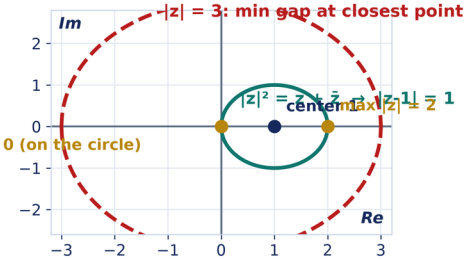
Worked — the disguised circle $|z|^2 = z + \bar{z}$

Write $z = x + iy$:
$x^2 + y^2 = 2x \Leftrightarrow (x-1)^2 + y^2 = 1$: circle of center 1, radius 1, through the origin. Then $\max|z|$ on it $= |1| + 1 = 2$ (farthest point: extend the ray $0 \to$ center by the radius) and $\min|z| = 0$ (the origin is on it).

$(z+1)(\bar{z}+1) = |z|^2 + z + \bar{z} + 1$-style expansions are the same move: expand, collect $z + \bar{z} = 2\mathrm{Re}\, z$, $z - \bar{z} = 2i\mathrm{Im}\, z$, $z\bar{z} = |z|^2$, then complete the square.

"Purely imaginary" loci: $\frac{z+1}{z-1}$ purely imaginary $\Leftrightarrow$ the fraction plus its conjugate is 0: numerator
$(z+1)(\bar{z}-1) + (\bar{z}+1)(z-1) = 2|z|^2 - 2 = ($
: the unit circle minus $z = 1$ (undefined there). Geometry confirms: $\arg\frac{z+1}{z-1} = \pm 90°$ is Thales' circle on diameter $[-1, 1]$ — the unit circle.

Disguised circle $|z|^2 = z + \bar{z}$: center 1, radius 1 — range of $|z|$ on it is $[0, 2]$



Ranges of $|z|$ over regions

Method: $|z|$ is the distance from 0; over a closed region the extreme distances are attained on the boundary (draw it, read them off). For open regions the range is open accordingly.

WORKED EXAMPLE

Worked — $|z+2| \leq 3$ **and the open-disk variant**

$|z+2| \leq 3$: closed disk center -2, radius 3. It contains 0 ($|0+2| = 2 \leq 3$): $\min|z| = 0$; farthest point from 0 is $-2-3 = -5$: $\max|z| = 5$. Range $[0, 5]$.
(Triangle-inequality check: $|z| \leq |z+2| + 2 \leq 5$, equality at $z = -5$ ✓.)

$\mathrm{Re}\, z > 0$ and $|z-1| < 1$: the open disk touches 0 (its leftmost point) and lies in $\mathrm{Re}\, z \geq 0$; the constraint removes just 0. Distances from 0 inside: $(0, 2)$ — open at both ends (the disk is open; its rightmost point 2 is excluded). Range $(0, 2)$.

STANDARD APPLICATION

Protocol for every locus problem

1. Translate the condition into $|z-a|$ -language (expand $z\bar{z}$, $z + \bar{z}$).
2. Identify the curve (circle / bisector / ellipse / line) **and its degenerate cases** ($k = 1$; ellipse constant $\leq$ focal distance).
3. Fix endpoints/exclusions (fractions: exclude poles; arcs: exclude endpoints).
4. If a max/min is asked: draw the region, mark 0 (or the target point), read nearest/farthest boundary points — then confirm with the triangle inequality and its equality condition.

COMMON TRAP

Common traps — modulus loci

- Center: $|z-a|=r$ has center at a; $|z+a|=r$ at $-a$.
- Degenerate cases first: $|z-1|+|z+1|=2$ is the **segment** $[-1,1]$, not an ellipse; $=1$ is the empty set; $|z-a|=|z-b|$ is a line, not a circle.
- Open vs closed: $|z|<1$ excludes the boundary — range answers must respect it ($(0,2)$, not $[0,2]$).
- In "purely imaginary" fraction problems, always exclude the pole ($z=1$ above) — substituting it back gives $\frac{2}{0}$.
- $\max|z|$ on a circle is center-distance + radius; $\min|z|$ is |center-distance − radius| — not the radius alone.

The inscribed-angle condition arg((z − a)/(z − b)) = θ

$\arg\frac{z-a}{z-b}$ is the directed angle from the ray $z\to b$ to the ray $z\to a$ (mod 2π). Fixing it at θ with $0<|\theta|<\pi$ says: **the chord** ab **subtends the constant angle** θ **at** z — by the inscribed-angle theorem, the locus is an **arc of a circle through** a **and** b (two arcs qualify; the sign/orientation of θ selects one).

KEY RESULT

Construction (memorise the two formulas)

For chord ab and angle θ with $0<|\theta|<\pi$:

$$R=\frac{|a-b|}{2\sin|\theta|},$$

center on the perpendicular bisector of ab

at distance $\frac{|a-b|}{2}\cot|\theta|$ from the midpoint

on the side that makes the inscribed angle $|\theta|$; the locus is the arc **opposite the center** (the major arc when $|\theta|<\frac{\pi}{2}$), excluding a and b themselves (the fraction is 0 or undefined there).

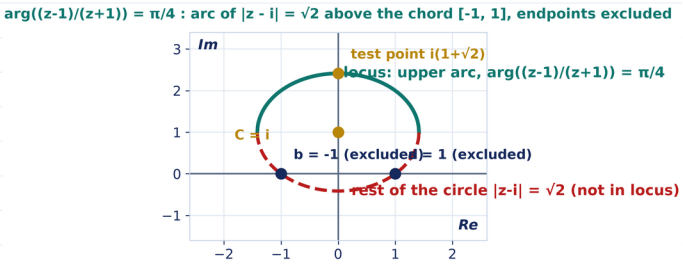
Ray version: $\arg(z-a)=\theta$ is the **ray** from a at angle θ (excluding a).

WORKED EXAMPLE

Worked — $\arg\frac{z-1}{z+1}=\frac{\pi}{4}$

$a=1$, $b=-1$; chord length 2; $R=\frac{2}{2\sin(\pi/4)}=\sqrt{2}$; midpoint 0; center on the imaginary axis at distance $\cot\frac{\pi}{4}=1$; center $C=i$ (the side is decided by testing a point). Test the top of the circle $z=i(1+\sqrt{2})$: $\arg(z-1)=\pi-\frac{3\pi}{8}=\frac{5\pi}{8}$, $\arg(z+1)=\frac{3\pi}{8}$, difference $\frac{\pi}{4}$ ✓ — so the locus is the **upper arc** of $x^2+(y-1)^2=2$ ($y>0$), endpoints ±1 excluded.

(Needed value: $\tan\frac{3\pi}{8}=\sqrt{2}+1$, the half-angle tangent of 45° — worth knowing.)



Ellipses — $|z-A|+|z-B|=2a_0$

Sum of distances to two fixed points constant $2a_0$: an **ellipse** with foci A, B when $2a_0 > |A-B|$; semi-major a_0, semi-minor $b_0 = \sqrt{a_0^2 - c^2}$ with $2c = |A-B|$. The **maximum of** $|\operatorname{Im} z|$ **is** b_0 (top of the minor axis) when the foci are real.

WORKED EXAMPLE

Worked — $|z-1|+|z+1|=6$; max of $\operatorname{Im} z$

Foci ± 1; $2c=2$, $c=1$; $2a_0=6$, $a_0=3$; $b_0=\sqrt{9-1}=2\sqrt{2}$. Max $\operatorname{Im} z = 2\sqrt{2}$ at $z=\pm i2\sqrt{2}$; check $|i2\sqrt{2}-1|=\sqrt{1+8}=3$, same for $+1$, sum 6 ✓.

Check the constant first:

$|z-1|+|z+1|=2$ is the segment $[-1,1]$ ($2a_0=2c$; degenerate); $=1$ is empty. The inequality $2a_0 > 2c$ must be verified before any ellipse work.

STANDARD APPLICATION

Argument vs modulus — the two-branch decision tree

Condition is a **modulus** of a linear fraction $=k$? Apollonius circle. **Modulus sum** $=$ const? ellipse (or segment/empty). **Argument** of a fraction $=$ const? arc of circle. **Arg of a linear term** $=$ const? ray. **Real/imaginary part** $=$ const? line. Mixed conditions ($|z-a|=r$ and $\arg z=\theta$)? Intersections — usually 0, 1 or 2 points: draw the circle and the ray, count.

COMMON TRAP

Common traps — argument loci

- Endpoints a, b are **never** on the locus (the fraction is 0 or ∞ there) — exclude them explicitly.
- The arc selection: test one point (as in the worked example) or track the sign of θ; the other arc gives $\arg = -\theta$ or $\pi - \theta$ -type values.
- $\arg w = \theta$ fixes w to one **ray**; $\arg w \in (\theta_1, \theta_2)$ is a sector; $\arg(w) = \theta$ or $\theta + \pi$ is a line through the origin in the w-plane — read the shape from the allowed set of values.
- Ellipse formula $b_0 = \sqrt{a_0^2 - c^2}$ needs $a_0 \geq c$; foci on the real axis is assumed for "$\max \operatorname{Im} z = b_0$" — for general foci rotate accordingly (or work with $|z|$-distances).
- $\arg \frac{z-a}{z-b}$ at z on the major arc vs minor arc: the two arcs see the chord under supplementary angles θ and $\pi - \theta$; only one matches the given directed value.

The three results, in increasing power

KEY RESULT

(A) Triangle inequality with equality condition

$$|z-a| \geq ||z| - |a||, \qquad |z-a| \leq |z| + |a|$$

Equality on the left iff z, a point the **same direction** (one is a non-negative real multiple of the other); on the right iff **opposite directions**. For constrained z this gives min/max in one line.

KEY RESULT

(B) Extremes of $|z-a|$ on the unit circle (and any circle)

If $|z|=r$ and $a\neq 0$:

$$\min|z-a| = ||a|-r| \quad \text{at } z = r\frac{a}{|a|}$$

$$\max|z-a| = |a|+r \quad \text{at } z = -r\frac{a}{|a|}$$

Geometric reading: nearest/farthest points of a circle to an exterior point lie on the line through the center and the point. For a circle $|z-z_0|=r$: replace $|z|$-distance by $|$distance from $z_0|$ — same formulas with $a \mapsto a-z_0$.

KEY RESULT

(C) Chord distance between unit points

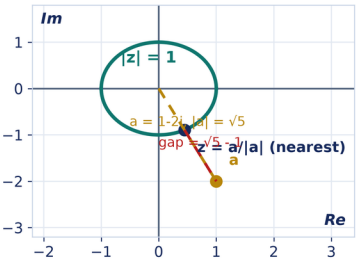
If $|z|=|w|=1$:

$$|z-w|^2 = (z-w)(\bar z-\bar w) = 2 - z\bar w - \bar z w$$

$$2 - 2\mathrm{Re}(z\bar w)$$

so $|z-w| = 2\left|\sin\frac{\varphi}{2}\right|$ where $\varphi = \arg(w/z)$ is the angle between them. In particular $|z-w|=\sqrt{2}$ iff $z\bar w = \pm i$ (quarter-turn apart); $|z-w|=1$ iff 60° apart ($z\bar w = e^{\pm i\pi/3}$).

min |z - a| on |z| = 1 : attained where the ray 0 → a exits the circle — gap √5 - 1



WORKED EXAMPLE

Worked — the standard three questions

1. $|z|=1$: $\min|z-(3+4i)| = 5-1 = 4$ at $z=\frac{3+4i}{5}$; $\max = 6$ at $z=-\frac{3+4i}{5}$.
2. $|z|=1$: range of $|z+\frac{1}{z}+2i|$: on the circle $\frac{1}{z}=\bar z$, so $z+\frac{1}{z}=2\cos\theta$ (real) and the expression $=2\cos\theta+2i$: modulus $2\sqrt{\cos^2\theta+1}\in[2,2\sqrt{2}]$. Min at $z=\pm i$, max at $z=\pm 1$.
3. $|z|=|w|=1$, $z\bar w = i$: $|z-w|^2 = 2-2\mathrm{Re}(i) = 2$; $|z-w|=\sqrt{2}$ ✓ (quarter-turn chord).

SHORT DERIVATION

Parallelogram-type minimisations — $|z-a|^2+|z-b|^2$

Expand $|z-a|^2+|z-b|^2 = 2|z-m|^2+\frac{1}{2}|a-b|^2$ with $m=\frac{a+b}{2}$ (complete the square in z; or read Leibniz's centroid identity of the olympiad sheets with $n=2$). So the unconstrained min is at the midpoint; under a constraint like $|z|=1$ the problem becomes "min $|z-m|$" = case (B) applied to m. Every "$\min(|z-a|^2+|z-b|^2)$ subject to..." question is this two-step.

STANDARD APPLICATION

Region questions — draw, then read distances

Range of $|z|$ (or $|z-a|$) for z in a region: nearest/farthest points are on the boundary — circle arcs give center ± radius; half-plane constraints clip candidates; ***vertices and tangency points win***. Always: (i) draw to scale, (ii) list candidate points (nearest/farthest on each boundary piece), (iii) evaluate, (iv) state open/closed as given. For complex objectives like $|z+\frac{1}{z}|$ substitute $z=e^{i\theta}$ first (unit circle only!) and reduce to a real-variable range.

COMMON TRAP

Common traps — optimisation

- Equality conditions, not just the inequality: $\min|z-a|$ on $|z|=r$ is $||a|-r|$ **attained at** $z=ra/|a|$ — forgetting the attaining z loses half the marks.
- $\min|z-a|+|z-b|$ unconstrained is $|a-b|$ (segment), not via circles; with $|z|=r$ it becomes an ellipse problem — check degeneracy first.
- $|z+\frac{1}{z}|$ simplifies only on $|z|=1$ (where $\frac{1}{z}=\bar{z}$); off the circle it is not $2|\cos\theta|$.
- Chord formula $|z-w|=2|\sin\frac{\varphi}{2}|$ needs **both** on the unit circle; for general radii use $|z-w|$ directly or scale first.
- Region ranges respect openness: open disk + strict inequalities $\Rightarrow$ open range endpoints.

The one-page register

KEY RESULT

Algebra (sheet 01)

$z=a+bi$; $\bar{z}=a-bi$; $z+\bar{z}=2\mathrm{Re}\,z$, $z-\bar{z}=2i\mathrm{Im}\,z$; $|z|^2=z\bar{z}$; $\frac{z}{w}=\frac{z\bar{w}}{|w|^2}$; $zw=0\Rightarrow z=0$ or $w=0$; i^n cycles with period 4. Conjugation: $\overline{z+w}=\bar{z}+\bar{w}$, $\overline{zw}=\bar{z}\bar{w}$, $\bar{\bar{z}}=z$.

KEY RESULT

Plane and polar (sheets 02–03)

$|z-w|$ = distance; midpoint $\frac{z_1+z_2}{2}$; triangle inequality $|z+w|\le|z|+|w|$ and reverse, with same-direction/opposite-direction equality conditions; parallelogram law $|z+w|^2+|z-w|^2=2|z|^2+2|w|^2$.

$\mathrm{Arg}\,z\in(-\pi,\pi]$; $z=r(\cos\theta+i\sin\theta)=re^{i\theta}$; products multiply moduli and **add** arguments; $\times i$ = quarter-turn CCW; $z+\bar{z}=2r\cos\theta$, $z-\bar{z}=2ir\sin\theta$; $|z|=1\Rightarrow\bar{z}=1/z$.

De Moivre: $(\cos\theta+i\sin\theta)^n=\cos n\theta+i\sin n\theta$ ($n\in\mathbb{Z}$); triple angles from $n=3$; $e^{i\pi}+1=0$.

KEY RESULT

Roots and roots of unity (sheets 04–06)

n-th roots of w: $R^{1/n}e^{i(\phi+2k\pi)/n}$, $k=0,\dots,n-1$ — regular n-gon; square-root formula $\pm(\sqrt{\frac{r+a}{2}}+i\,\mathrm{sgn}(b)\sqrt{\frac{r-a}{2}})$.

$\omega=e^{2\pi i/3}$: $1+\omega+\omega^2=0$, $\bar{\omega}=\omega^2$, $(1-\omega)(1-\omega^2)=3$, $1+\omega=e^{i\pi/3}$, $1-\omega=\sqrt{3}e^{-i\pi/6}$.

Vanishing sum: $\sum_{k=0}^{n-1}\zeta^{mk}=n$ if $n\mid m$, else 0.

Products: $\prod_{k=1}^{n-1}(1-\zeta_k)=n$, $\prod_{k=1}^{n-1}\sin\frac{\pi k}{n}=\frac{n}{2^{n-1}}$, $\prod_{k=0}^{n-1}|z-\zeta_k|=|z^n-1|$. Filter: $\sum_{k\equiv r\,(\mathrm{mod}\,n)}\binom{N}{k}=\frac{1}{n}\sum_j e^{-2\pi irj/n}(1+e^{2\pi ij/n})^N$.

KEY RESULT

Equations, loci, optimisation (sheets 07–10)

Linear in $z,\bar{z}$: real/imag parts, or conjugate + 2×2. Real quadratics, $\Delta<0$: $|z_1-z_2|=\sqrt{|\Delta|}$, roots conjugate.

Loci: circle $|z-z_0|=r$; bisector $|z-a|=|z-b|$; Apollonius $|z-a|=k|z-b|$; ellipse $|z-A|+|z-B|=2a_0$ (max height $b_0=\sqrt{a_0^2-c^2}$); arc: $\arg\frac{z-a}{z-b}=\theta$, $R=\frac{|a-b|}{2\sin|\theta|}$, center on bisector at distance $\frac{|a-b|}{2}\cot|\theta|$, endpoints excluded.

Optimisation: $\min_{|z|=r}|z-a|=||a|-r|$ at $z=ra/|a|$; max $=|a|+r$; chord $|z-w|=2|\sin\frac{\varphi}{2}|$ for unit points; $|z-a|^2+|z-b|^2=2|z-m|^2+\frac{1}{2}|a-b|^2$.

Strategy card — the order that avoids every trap

STANDARD APPLICATION

Five steps for any complex-number problem

1. **Translate**: what is the algebraic shape ($z\bar{z}$, $|z-a|$, arg) and the geometric shape (distance / rotation / locus)?
2. **Choose the language**: compute in algebra, see structure in geometry; solve in whichever is transparent, verify in the other when possible.
3. **Check degenerate cases**: $k=1$ Apollonius, ellipse constant $\le$ focal distance, $z=0$, $w=0$, $|z|=1$ vs not.
4. **Track the branch/window**: $\mathrm{Arg} \in (-\pi, \pi]$; principal square root; equality conditions of the triangle inequality.
5. **Verify numerically**: plug the answer back ($z=\frac{8}{3}-\frac{i}{3}$ style) or compute both sides — one number check kills most silent errors.

COMMON TRAP

Final trap sweep

1. $-2i^2 = +2$; $\mathrm{Im}(3+4i) = 4$.
2. $\mathrm{Arg}(-1) = \pi$; fourth-quadrant arguments negative; Arg is not additive.
3. "Find the cube roots" = all three; $\sqrt{z}\sqrt{w} \neq \sqrt{zw}$.
4. $(1+\omega)(1+\omega^2) = 1$; $\sum_{k=0}^{2023} \omega^k = 1+\omega$.
5. $|z-a| = r$ center at a; degenerate locus cases checked first; endpoints excluded on arcs.
6. Equality conditions stated; region ranges respect open/closed.
7. $\bar{z} = 1/z$ only on $|z|=1$; $z+\frac{1}{z} = 2\cos\theta$ only there.

Where this toolkit reappears

- **Trigonometry (ch 3)**: De Moivre re-derives multiple angles; $\prod \sin\frac{\pi k}{n}$ matches the olympiad sine products; $\cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2}$ is the bridge.
- **Quadratic equations / sequences (ch 6–7)**: conjugate-root symmetry, Vieta, geometric series on the unit circle.
- **Coordinate geometry & vectors (ch 12–16)**: rotation formula $X' = A + e^{i\theta}(X-A)$ is the vector rotation; Ptolemy and distance identities become coordinate proofs.
- **Calculus (ch 17–21)**: $\tan^{-1}x = \arg(1+xi)$ -style links; limits of $(\cos\frac{\theta}{n} + i\sin\frac{\theta}{n})^n$ foreshadow $e^{i\theta}$.
- **Olympiad Part II of this chapter**: geometry-by-complex-bash, Gauss sums, Gauss–Lucas, Pompeiu.

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/geometry-via-complex extension (IOQM/RMO core weapon)

Encoding a figure's vertices as complex numbers and proving theorems by algebra is standard olympiad technique (and an occasional JEE Advanced strand). Everything here is derived from Part I (rotation = $\times e^{i\theta}$, ω-identities). The equilateral condition appears in JEE-Advanced-flavoured problems regularly; Ptolemy/Napoleon/van Aubel (next sheets) are classical olympiad theorems. No claim that these "can never appear" in JEE — they are simply at the harder, technique-heavy end.

The engine: rotation is multiplication

KEY RESULT

Rotation by θ about A

$$X' = A + e^{i\theta}(X - A)$$

One line: $X - A$ is the vector $A \to X$; multiplying by $e^{i\theta}$ keeps its length and adds θ to its direction; adding A re-anchors. Quarter turns: $\times i$; half turn: $\times(-1)$; 60°: $\times e^{i\pi/3} = \times(1+\omega)$ (identity 4 of sheet 05) — and $e^{-i\pi/3} = -\omega = \overline{1+\omega}$.

KEY RESULT

Equilateral-triangle condition (orientation-sensitive)

*(a, b, c) is equilateral **counterclockwise** $\Leftrightarrow$*

$$a + \omega b + \omega^2 c = 0$$

(clockwise ordering gives $a + \omega^2 b + \omega c = 0$). Both directions are one-line proofs below; the condition is the single most-used complex-geometry fact in olympiads.

SHORT DERIVATION

Proof both directions

($\Rightarrow$) CCW equilateral: $c - a = e^{i\pi/3}(b-a) = (1+\omega)(b-a)$ *(rotate side ab by $+60^\circ$). Solve:* $c = -\omega a + (1+\omega)b$. *Substitute:* $a + \omega b + \omega^2 c = a + \omega b - \omega^3 a + (\omega^2 + \omega^3)b$ *using* $\omega^3 = 1$, $\omega^2 + 1 = -\omega$. ∎

($\Leftarrow$) From $a + \omega b + \omega^2 c = 0$: $c = -\omega a - \omega^2 b$ *(using* $\omega^{-2} = \omega$*), so* $c - a = -(1+\omega)a - \omega^2 b = (1+\omega)(b-a)$ *(expand and use* $1 + \omega + \omega^2 = 0$*). Hence* $|c-a| = |b-a|$ *with a* 60° *turn: equilateral, CCW.* ∎

Check on $a = 0$, $b = 1$, $c = e^{i\pi/3}$: $\omega + \omega^2(1+\omega) = \omega + \omega^2 + \omega^3 = 0$ ✓.

STANDARD APPLICATION

Standard tests (translation dictionary)

Collinear z_1, z_2, z_3: $\frac{z_1 - z_2}{z_1 - z_3} \in \mathbb{R}$ *(the ratio of two side-vectors is real). Perpendicular at* z_1: *that ratio is purely imaginary* ($\in i\mathbb{R}$). *Equal lengths:* $|z_1 - z_2| = |z_3 - z_4|$. *Section formula: point dividing* $z_1 z_2$ *in ratio* $m:n$ *is* $\frac{nz_1 + mz_2}{m+n}$. *Angle* θ *at* z *between* a, b: $\arg\frac{b-z}{a-z} = \pm\theta$ *— the sheet-09 arc language. Concyclic: cross-ratio* $\frac{(z_1-z_3)(z_2-z_4)}{(z_1-z_4)(z_2-z_3)} \in \mathbb{R}$ *(olympiad-level; use only when needed).*

KEY RESULT

Area and circumcenter — two more one-liners

$$\text{Area}(\triangle z_1z_2z_3) = \tfrac{1}{2}\left|\operatorname{Im}\left((z_2-z_1)\overline{(z_3-z_1)}\right)\right|$$

Reason: $(z_2-z_1)\overline{(z_3-z_1)}$ is the cross product of the two sides in complex clothing — modulus |product of lengths|, argument = signed angle between them; its imaginary part is the signed twice-area. Shoelace is this formula in coordinates.

Cumcenter u: solve $|u-a|^2 = |u-b|^2 = |u-c|^2$: expanding, $u(\bar{b}-\bar{a}) + \bar{u}(b-a) = |b|^2 - |a|^2$ — one linear equation in $u, \bar{u}$ per pair; two of them, 2×2 solve. No perpendicular bisectors drawn.

WORKED EXAMPLE

Worked — area of $\triangle(1, 3i, -2+2i)$; **circumcenter of** $(1, i, -1)$

$z_2 - z_1 = -1+3i$, $z_3 - z_1 = -3+2i$; $(z_2-z_1)\overline{(z_3-z_1)} = (-1+3i)(-3-2i) = 3+$. Area $= \frac{1}{2}|-7| = \frac{7}{2}$. (Shoelace check: $\frac{1}{2}|1\cdot(3-2)+0+(-2)(0-3)| = \frac{1}{2}\cdot 7$ ✓.)

Circumcenter of $1, i, -1$: all on $|z|=1$, so $u = 0$ (uniqueness of the circumcenter of a non-degenerate triangle). The linear solve confirms: $u(-i-1) + \bar{u}(i-1) = |i|^2 - |1|^2 = 0$, satisfied by $u = 0$ ✓.

COMMON TRAP

Traps — rotation toolkit

- Orientation: $a+\omega b+\omega^2 c = 0$ is **CCW-only**; a problem's clockwise labelling needs $\omega \leftrightarrow \omega^2$. If ordering is unstated, check both or phrase conclusions orientation-free.
- $\frac{z_1-z_2}{z_1-z_3} \in \mathbb{R}$ excludes $z_1 = z_3$ (division) — degenerate configurations need separate handling.
- Area formula has a $\frac{1}{2}$ and an absolute value; the signed imaginary part tracks orientation.
- Circumcenter 2×2 solve fails when $a = b$ (degenerate); assume distinct vertices.
- $e^{i\pi/3} = 1+\omega$ and $e^{-i\pi/3} = -\omega$ — mixing these up reverses every rotation in sight.

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad extension (classical geometry ↔ algebra)

Ptolemy's theorem/inequality and distance theorems (British Flag and friends) are proved here as pure algebra in $\mathbb{C}$. The identity behind Ptolemy is a 30-second expansion usable in any exam; the equality-condition analysis is olympiad-grade. The JEE-visible content: the identity, the inequality, and the distance expansions — all derivable from Part I.

The Ptolemy identity — no geometry anywhere

KEY RESULT

Identity (valid for ALL four complex numbers a, b, c, d**)**

$$(a-c)(b-d) = (a-b)(c-d) + (b-c)(a-d)$$

Verify yourself: RHS $= (ac-ad-bc+bd) + (ab-bd-ca+cd)$ ✓. (Expansion only — no circle, no ordering, nothing.)

SHORT DERIVATION

Ptolemy's inequality and theorem — one modulus step each

Take moduli and use the triangle inequality:

$$|a-c|\,|b-d|$$

$$|(a-b)(c-d)+(b-c)(a-d)|$$

$$\le |a-b|\,|c-d|+|b-c|\,|a-d|$$

— the product of the diagonals is at most the sum of products of opposite sides, for **any** four points. Equality holds iff $(a-b)(c-d)$ and $(b-c)(a-d)$ point the same direction, i.e. $\frac{(b-c)(a-d)}{(a-b)(c-d)} \in \mathbb{R}_{>0}$; taking arguments this says the angles the chord bd subtends at a and at c are **supplementary** — the points are concyclic in the order a, b, c, d (inscribed-angle theorem, oly-01's collinearity language). That is Ptolemy's theorem: for a cyclic quadrilateral, equality — the diagonal-product formula from school geometry. ∎

Unit-square check ($a=1, b=i, c=-1, d=-i$): diagonals $2\times 2=4$; sides $\sqrt{2}$: $\sqrt{2}\cdot\sqrt{2}+\sqrt{2}\cdot\sqrt{2}=4$ ✓.

STANDARD APPLICATION

How to use Ptolemy in an olympiad

1. Encode the four points; write the identity (or just take the cyclic special case if cyclicity is given).
2. For length-product equalities/inequalities ($PA\cdot QC+\ldots$), match to the four factors.
3. Collinearity-ish corollaries: if one "vertex" collapses onto a side, Ptolemy degenerates to the triangle inequality $|a-b|\le|a-c|+|c-b|$ — recognise the degeneration instead of re-deriving.
4. The converse direction (equality ⇒ cyclic) is available when a problem says "prove $ABCD$ cyclic" via lengths.

Distance theorems as expansions — the British Flag

KEY RESULT

British Flag theorem (rectangle form)

For a rectangle A, B, C, D (diagonal pairs A, C and B, D) and **any** point e in the plane:

$$|e-A|^2+|e-C|^2=|e-B|^2+|e-D|^2$$

SHORT DERIVATION

Proof — expand $|z-a|^2=|z|^2-z\bar{a}-\bar{z}a+|a|^2$ and collect

Take the square with vertices $0, 1, 1+i, i$; the statement is $|e|^2+|e-(1+i)|^2=|e-1|^2+|e-i|^2$. Expand each:

$$|e-(1+i)|^2=|e|^2-e(1-i)-\bar{e}(1+i)+2,$$

$$|e-1|^2=|e|^2-e-\bar{e}+1,$$

$$|e-i|^2=|e|^2+ie-i\bar{e}+1$$

LHS $=2|e|^2-e+ie-\bar{e}-i\bar{e}+2$; RHS $=2|e|^2-e+ie-\bar{e}-i\bar{e}+2$ — identical term for term. ∎ (For a general rectangle with vertices $0, w, w+v, v$ the same expansion cancels because the vertex constants pair up: $0+(w+v)=w+v$.)

The pattern worth internalising: $|z-a|^2$ **is a quadratic in $z, \bar{z}$ with clean coefficients** — every rectangle/square distance theorem is "expand and collect".

WORKED EXAMPLE

Worked — the same expansion proves the parallelogram law and the Leibniz $n=2$ case

$|z+w|^2+|z-w|^2$: expand $=(|z|^2+|w|^2+z\bar{w}+\bar{z}w)+(|z|^2+|w|^2-z\bar{w}$ ✓ (sheet 01). And $|z-a|^2+|z-b|^2=2|z-m|^2+\frac{1}{2}|a-b|^2$ with $m=\frac{a+b}{2}$: expand both sides, compare (or complete the square). Three "theorems", one algebraic habit.

COMMON TRAP

Traps — Ptolemy and expansions

- The identity is unconditional; the **theorem** (equality) needs cyclic order — never claim $|a-c||b-d| = \dots$ for a generic quadrilateral.
- Ptolemy's factors pair as (diagonal, diagonal) vs (opposite side, opposite side) — the ordering a, b, c, d around the circle is part of the statement.
- British Flag is for **rectangles** (points in the plane, any point e — inside or outside); for general quadrilaterals it fails.
- When expanding $|z-a|^2$, note $\overline{(z-a)} = \bar{z} - \bar{a}$ — conjugating the constant correctly is the usual slip.
- Degenerate Ptolemy (a point on a side) gives the triangle inequality — good consistency check, not a new theorem.

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad extension (classical theorems, JEE-adjacent technique)

Van Aubel (squares on the sides of a quadrilateral) and Napoleon (equilateral triangles on the sides of a triangle) are olympiad staples proved here by the ω-machinery of Part I. The technique (center formulas + the equilateral condition) is exactly the JEE-Advanced "geometry via complex numbers" strand; the theorems themselves sit at the IOQM/RMO end. Derivable from Part I + oly-01; nothing assumed beyond them.

Outward-square centers

KEY RESULT

Center formula (counterclockwise quadrilateral, outward square on side pq)

$$m_{pq} = \frac{p+q}{2} - \frac{i}{2}(q-p) = \frac{(1+i)p + (1-i)q}{2}$$

Reason: the center of any square on base pq is the midpoint $\pm$ the half-side rotated by 90°; for a CCW quadrilateral the interior lies to the left of $p \to q$, so the outward direction is $-i(q-p)$'s side. (Check on side $0 \to 1$: center $\frac{1-i}{2}$ — below the real axis ✓.)

SHORT DERIVATION

Van Aubel — opposite centers are equal and perpendicular

For CCW a, b, c, d with outward centers $m_{ab}, m_{bc}, m_{cd}, m_{da}$:

$$2(m_{bc} - m_{da})$$

$$((1+i)b + (1-i)c) - ((1+i)d + (1-i)a)$$

$$2(m_{cd} - m_{ab})$$

$$((1+i)c + (1-i)d) - ((1+i)a + (1-i)b)$$

Multiply the second by i and collect: $i \cdot 2(m_{cd} - m_{ab}) = -2(m_{bc} - m_{da})$, i.e.

$$m_{bc} - m_{da} = -i\,(m_{cd} - m_{ab}).$$

One equation, two facts: $\times(-i)$ is a 90° rotation preserving length — the segments are ***equal and perpendicular***. ∎ (Numeric check on $0, 1, 1.5+1.2i, 0.3+0.9i$: both segments ≈ 2.1645 long, ratio pure imaginary ✓.)

Napoleon's Theorem — equilaterals erected on the sides

KEY RESULT

Outward-equilateral center (CCW triangle a, b, c, third vertex via -60° turn)

Third vertex of the outward equilateral on $a \to b$: $a + e^{-i\pi/3}(b-a) = a - \omega(b-a)$. Its center:

$$m_{ab} = \frac{a + b + (a - \omega(b-a))}{3} = \frac{(2+\omega)a + (1-\omega)b}{3}$$

using $e^{-i\pi/3} = -\omega$ and $1 - \omega^2 = 2 + \omega$.

SHORT DERIVATION

Napoleon — the centers form an equilateral triangle

By oly-01's condition, suffices to show $m_{ab} + \omega m_{bc} + \omega^2 m_{ca} = 0$. Compute $3 \times$ the left side and collect the a-coefficient:

$$(2+\omega) + \omega^2(1-\omega) = 2 + \omega + \omega^2 - \omega^3 = 1 + \omega + \omega^2$$

$$0$$

and identically for b and c (each coefficient collapses to $1 + \omega + \omega^2 = 0$). Hence $m_{ab} + \omega m_{bc} + \omega^2 m_{ca} = 0$: the outer Napoleon triangle is equilateral. ∎

Numeric check on $(0,0), (4,0), (1,3)$: centers $(2, -1.1547)$, $(3.366, 2.366)$, $(-0.366, 1.789)$; all three mutual distances $\sqrt{14.261}$ ✓.

STANDARD APPLICATION

Technique takeaway — center formulas are linears in the vertices

Any "centers of figures erected on the sides" problem is: (i) write each center as an affine combination of its two base vertices (fixed complex coefficients depending only on the shape erected); (ii) form the target combination; (iii) the vertex coefficients must vanish (or match) — this *is* the theorem. Works for squares (van Aubel), equilaterals (Napoleon), similar triangles, and regular n-gons. The coefficients come from the rotation factors $e^{i\theta}$ of oly-01.

COMMON TRAP

Traps — outward constructions

- Outward vs inward: the sign in $\frac{p+q}{2} \pm \frac{i}{2}(q-p)$ flips the construction; "outward" is fixed by the polygon's orientation (CCW assumed throughout these sheets). Inward squares give a companion theorem (segments still equal/perpendicular) — same proof, sign flip.
- $\omega = e^{2\pi i/3}$ in the equilateral condition, but the 60° turn of a side is $e^{\pm i\pi/3}$: $e^{-i\pi/3} = -\omega$, $e^{i\pi/3} = 1 + \omega$ — do not conflate the ω of the condition with the rotation factor.
- Van Aubel is for **any** quadrilateral (convex or not, with outward defined by consistent orientation); Napoleon for any triangle.
- "Centers" are centroids (of the square/equilateral), not vertices or circumcenters — the coefficients above are centroid-specific ($\frac{1}{3}$ for triangles, $\frac{1}{2} + \frac{i}{2}$-type for squares).

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad extension (synthesis problems)

Three advanced-but-derivable strands: the Leibniz centroid distance formula (uses only $|z-a|^2$ expansions), unit-circle configurations where $|z_k| = 1$ forces a shape (uses $\bar{z} = 1/z$), and the map $w = z + \frac{1}{z}$ (Joukowski) collapsing circles to segments. All appear in IOQM/RMO and in coaching "advanced complex numbers" sets; all are derivable from Part I.

Leibniz — distance sums are minimised at the centroid

KEY RESULT

Centroid identity (points $v_1, \dots, v_n$, centroid $c = \frac{1}{n}\sum v_k$, any p)

$$\sum_{k=1}^{n} |p - v_k|^2 = n\,|p - c|^2 + \sum_{k=1}^{n} |v_k - c|^2$$

SHORT DERIVATION

Proof — expand $p - v_k = (p - c) - (v_k - c)$

$$\sum |p - v_k|^2$$

$$= \sum |(p - c) - (v_k - c)|^2$$

$$= n|p - c|^2 + \sum |v_k - c|^2 - 2\mathrm{Re}\left((p - c)\overline{\sum (v_k - c)}\right)$$

and $\sum (v_k - c) = \sum v_k - nc = 0$ kills the cross term. ∎ Reading: the second term is the fixed "inertia" of the configuration; the first is minimised at $p = c$ — **the centroid is the point of minimum total squared distance**. Corollary for a regular n-gon of circumradius R centered at 0 with $|p| = 1$: sum $= n \cdot 1 + nR^2$ — e.g. hexagon $R = 2$, $|P| = 1$: $6 + 24 = 30$ (independent of the direction of P).

Unit-circle configurations — $\bar{z} = 1/z$ is the entry ticket

STANDARD APPLICATION

Four unit numbers summing to zero form a rectangle

If $|z_k| = 1$ and $z_1 + z_2 + z_3 + z_4 = 0$; conjugate the sum and use $\bar{z}_k = 1/z_k$:

$$\frac{1}{z_1} + \frac{1}{z_2}$$

$$= -\left(\frac{1}{z_3} + \frac{1}{z_4}\right) \Rightarrow (z_1 + z_2)\left(\frac{1}{z_1 z_2} - \frac{1}{z_3 z_4}\right)$$

$$= 0$$

(using $z_3 + z_4 = -(z_1 + z_2)$). Either $z_1 + z_2 = 0$ (antipodal pairs $\Rightarrow \{\pm a, \pm b\}$: a cyclic parallelogram = rectangle), or $z_1 z_2 = z_3 z_4$, which makes $\{z_3, z_4\} = \{-z_1, -z_2\}$ (same shape: $t^2 + st + p = 0$ has roots $-1 \times$ those of $t^2 - st + p = 0$). Square exactly when adjacent vertices are 90° apart ($z_2 = \pm i z_1$). ∎

STANDARD APPLICATION

Three unit numbers summing to zero are 120° apart

$|z_1 + z_2| = |z_3| = 1 \Rightarrow 1 = 2 + 2\mathrm{Re}(z_1 \bar{z}_2) \Rightarrow \mathrm{Re}$ mutual angles 120°: an equilateral triangle centered at 0; the six points $\{\pm z_k\}$ step by 60°: a ***regular hexagon***. (Check: antipodes land exactly halfway between the 120° -spaced points.)

KEY RESULT

Product identity — distances to the n-th roots

$$\prod_{k=0}^{n-1} |z - \zeta_k| = |z^n - 1|$$

(factor $z^n - 1 = \prod (z - \zeta_k)$, take moduli). E.g. vertices of the regular pentagon on $|z| = 1$, distances from $z = 2$: $\prod |2 - \zeta_k| = |32 - 1| = 31$. On the circle itself ($z = e^{i\theta}$): $\prod_k |e^{i\theta} - \zeta_k| = 2\left|\sin \frac{n\theta}{2}\right|$.

Joukowski — $w = z + 1/z$ collapses circles to segments

On $|z| = 1$: $\frac{1}{z} = \bar{z}$, so $w = z + \bar{z} = 2\cos\theta \in [-2, 2]$: the unit circle maps ***two-to-one*** onto the real segment $[-2, 2]$ (once each at ± 1). Affine images: $w = z + \frac{1}{z} + i$ maps the circle to the segment $[-2 + i, 2 + i]$ — translate the segment by i. Off the circle: $|z| = r \neq 1$ gives the ellipse $\frac{u^2}{(r + 1/r)^2} + \frac{v^2}{(r - 1/r)^2} = 1$ in $w = u + iv$ (derive: $w = re^{i\theta} + r^{-1}e^{-i\theta} = (r + r^{-1})\cos\theta + i(r - r^{-1})$si) — circles become ellipses, degenerating to the segment at $r = 1$.

WORKED EXAMPLE

Worked — locus of $w = z + \frac{1}{z} + i$ as $|z| = 1$

$w = 2\cos\theta + i$: the horizontal segment from $-2 + i$ to $2 + i$, traced twice (endpoints once). Every point attained ($\cos\theta = \frac{\mathrm{Re}\, w}{2}$). For range-type JEE questions ("find the range of $|z + \frac{1}{z} + 2i|$ on $|z| = 1$") the same reduction is sheet-10 method: $2\cos\theta + 2i \Rightarrow [2, 2\sqrt{2}]$ ✓ consistent.

COMMON TRAP

Traps — synthesis

- $\bar{z} = 1/z$ requires $|z| = 1$; conjugating a relation to introduce $1/z$ is illegal off the circle.
- Rectangle conclusion from four-unit-sum-zero assumes all four on the unit circle; drop that and the statement dies (z_k arbitrary with zero sum says nothing).
- Joukowski collapse is **two-to-one** (z and $1/z$ map to the same w); locus answers list the image set, not parametrisations.
- The product identity's index runs 0 to $n - 1$ (includes $z = 1$'s factor $|z - 1|$); the $\prod(1 - \zeta_k) = n$ formula of sheet 06 is its $z = 1$ cousin **excluding** ζ_0 — keep the two consistent (at $z = 1$: $|z^n - 1| = 0$; the nonzero cousin excludes $k = 0$).
- Leibniz needs the **centroid**; using the circumcenter instead gives the wrong "minimum" statement (the circumcenter minimises max distance, a different problem).

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad extension (deeper number theory; IOQM/RMO+)

The quadratic Gauss sum $G(n) = \sum_{k=0}^{n-1} \zeta^{k^2}$ is classical olympiad/number-theory material (the analytic heart of quadratic reciprocity). Its **evaluation mechanism** — square, re-index, let the vanishing sum filter — uses nothing but Part I sheets 05–06. The full reciprocity connection is quoted as context, not required. This is specialised extension material, not core JEE; nothing prevents an exam from asking the small cases ($n = 3, 4, 5$) computationally.

The object and its closed form

For $\zeta = e^{2\pi i/n}$ define $G(n) = \sum_{k=0}^{n-1} \zeta^{k^2}$ (the exponent is k^2 — this is why the vanishing-sum filter does not kill it outright).

KEY RESULT

Closed form (phase locked to n mod 4)

$G(n) = \sqrt{n}$ if $n \equiv 1\ (4)$; $= i\sqrt{n}$ if $n \equiv 3\ (4)$

$G(n) = (1 + i)\sqrt{n}$ if $n \equiv 0\ (4)$; $= 0$ if $n \equiv 2\ (4)$

Mechanism 1: square, then re-index

Mechanism 1: square, then re-index

SHORT DERIVATION

$|G(n)|^2$ — the double sum the filter can collapse

Each term has modulus 1, so $|G(n)|^2 = \sum_{k,l} \zeta^{k^2-l^2}$. Put $k = l + h$: $k^2 - l^2 = 2lh + h^2$, and

$$|G(n)|^2 = \sum_{h=0}^{n-1} \zeta^{h^2} \sum_{l=0}^{n-1} \zeta^{2lh}$$

- n odd: $n \mid 2h \Rightarrow n \mid h$: only $h = 0$ survives: $|G(n)|^2 = n$.
- $n = 2m$: $n \mid 2h \Leftrightarrow m \mid h$: $h = 0$ and $h = m$ survive; $\zeta^{m^2} = e^{\pi i m} = (-1)^m$, so $|G(n)|^2 = n(1 + (-1)^m)$: 0 for m odd ($n \equiv 2 \pmod 4$), $2n$ for m even ($n \equiv 0 \pmod 4$).

Squaring is the whole trick: it turns a 1-dimensional sum into a 2-dimensional one the filter collapses. ∎

Mechanism 2: the phase, case by case

Since $k^2 \equiv (n-k)^2 \pmod n$, terms pair: $G(n) = 1 + 2\sum_{k=1}^{(n-1)/2} \zeta^{k^2}$ for odd n.

WORKED EXAMPLE

Small cases, computed exactly

$G(3) = 1 + 2\omega = 1 + 2(-\frac{1}{2} + i\frac{\sqrt{3}}{2}) = i\sqrt{3}$ ✓ ($n = 3 \equiv 3$: phase i).

$G(5) = 1 + 2(\zeta + \zeta^4) = 1 + 4\cos\frac{2\pi}{5} = 1 + (\sqrt{5}$ ✓ ($n \equiv 1$: phase 1), using $\cos\frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$.

$G(7) = 1 + 2(\zeta + \zeta^2 + \zeta^4) = i\sqrt{7}$ ✓ ($n = 7 \equiv 3$); the exponent residues $k^2 \pmod 7$ run $0, 1, 4, 2, 2, 4, 1$.

$G(8)$: residues $k^2 \pmod 8$ are $0, 1, 4$ with 4 twice and 1 four times: $G(8) = 2(1 + \zeta^4) + 4\zeta = 4 \cdot \frac{1+i}{\sqrt{2}} = (1+i)\sqrt{8}$ ✓ ($n \equiv 0$).

$G(2)$: here $\zeta = e^{2\pi i/2} = -1$ and $k = 0, 1$ give $\zeta^0 + \zeta^1 = 1 + (-1) = 0$ ✓ (vanishes, as the mod-4 rule predicts). Likewise $G(6) = G(10) = 0$.

STANDARD APPLICATION

How to use this in a problem

1. **Vanishing case first**: if $n \equiv 2 \pmod 4$, the answer is 0 before any arithmetic.
2. Odd n: pair k with $n - k$, evaluate the half-sum in trig ($\cos\frac{2\pi k}{n}$ values) when n is small; or quote the phase rule with the modulus $\sqrt{n}$.
3. The Legendre-symbol form $G(p) = \sum_{k=0}^{p-1} (\frac{k}{p})\zeta^k$ for odd primes p is the same number; $G(p) = \sqrt{p}$ or $i\sqrt{p}$ by $p \pmod 4$ is quadratic reciprocity's analytic core — quote as context.
4. Verification habit: compute $|G(n)|^2$ numerically (should be n, 0, or $2n$ per the table) — one number confirms the phase branch.

COMMON TRAP

Traps — Gauss sums

- $n \equiv 2 \pmod 4$: the sum is exactly 0 — not "small"; the two surviving filter contributions cancel (1 and $(-1)^m$ with m odd).
- The phase: $n \equiv 1$: real positive; $n \equiv 3$: purely imaginary positive; $n \equiv 0$: $(1+i)\sqrt{n}$ (argument $\frac{\pi}{4}$) — mixing up $n \pmod 4$ with $n \pmod 3$ because "odd/even" feels like enough is the standard slip.
- The exponent is k^2 **mod** n inside ζ^{k^2} only implicitly: $\zeta^{k^2} = \zeta^{k^2 \pmod n}$ — when pairing residues, reduce mod n first.
- $G(2) = 0$ uses $\zeta = e^{2\pi i/2} = -1$; reading ζ as $i = e^{2\pi i/4}$ (the $n = 4$ root) gives the wrong sum $1 + i$. ζ is always $e^{2\pi i/n}$ for the n of the problem.
- ζ depends on n; every line must use the same n throughout.

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (deeper theory; IOQM/RMO+)

Three structure theorems about polynomials and equilateral triangles. **Gauss–Lucas** (critical points stay in the convex hull of the roots) is the complex substitute for Rolle's theorem; **Marden** pinpoints the critical points of a cubic as ellipse foci; **Pompeiu** says PA, PB, PC from an equilateral triangle always form a triangle. Each proof here is a few lines of Part I algebra; the theorems themselves are beyond the core JEE syllabus sentence and are labelled accordingly. No claim that any of this "can never appear" in JEE Advanced — the computational corollaries (critical points of a cubic; distance-triangle inequalities) are ordinary Part I manipulations.

Gauss–Lucas — critical points cannot leave the hull

KEY RESULT

Theorem

Let $p(z) = c\prod_{k=1}^{n}(z - r_k)$. Then every zero of $p'(z)$ lies in the **convex hull** of $\{r_1, \dots, r_n\}$.

SHORT DERIVATION

Proof — the logarithmic derivative is a sum of point charges

Logarithmic differentiation: $\frac{p'(z)}{p(z)} = \sum_{k=1}^{n}\frac{1}{z - r_k}$. If z lies outside the hull, a line separates it from all roots: for some real θ, $\operatorname{Re}\left(e^{-i\theta}(z - r_k)\right) > 0$ for all k. Then

$$\operatorname{Re}\left(e^{i\theta}\frac{p'(z)}{p(z)}\right)$$

$$= \sum_{k=1}^{n}\frac{\operatorname{Re}\left(e^{-i\theta}(z - r_k)\right)}{|z - r_k|^2} > 0$$

so $p'(z)/p(z) \neq 0$, hence $p'(z) \neq 0$ — contradiction. ∎

Corollary (Rolle in complex form): all roots real ⇒ hull is a segment of the real line ⇒ every critical point is real — no intermediate-value argument anywhere.

WORKED EXAMPLE

Worked — critical points of $p(z) = z(z-2)(z-1-2i)$

Expand: $p(z) = z^3 + (-3-2i)z^2 + (2+4i)z$; $p'(z) = 3z^2 + (-6-4i)z + (2+4i)$.

Discriminant: $b^2 - 4ac = (20+48i) - (24+48i) = -4$, so $\sqrt{\text{disc}} = \pm 2i$:

$$z = \frac{6+4i \pm 2i}{6} \in \left\{1+i,\ 1+\tfrac{i}{3}\right\}.$$

Both lie inside the triangle $(0, 2, 1+2i)$ (half-plane check: $y \geq 0$, $y \leq 2x$, $y \leq 4-2x$ ✓), exactly as Gauss–Lucas demands. Shifting the centroid $1+\frac{2}{3}i$ to 0 sends the two points to $\pm\frac{i}{3}$ — symmetric about the centroid, which is Marden's signature.

STANDARD APPLICATION

Marden's theorem (quote-level)

For a cubic whose roots are the vertices of a triangle, the two roots of p' are the ***foci of the Steiner inellipse*** (the ellipse tangent to the three sides at their midpoints). Gauss–Lucas puts them inside the triangle; Marden identifies them exactly. Test with distances: each side midpoint m_k satisfies $|m_k - f_1| + |m_k - f_2| =$ the same constant. For an equilateral triangle the foci coincide at the center and p' has a double root there.

Pompeiu — three distances that always make a triangle

KEY RESULT

Theorem

ABC equilateral, P any point of the plane: PA, PB, PC are the side lengths of a (possibly degenerate) triangle; degenerate ***exactly*** when P lies on the circumcircle of ABC.

SHORT DERIVATION

Proof — the equilateral condition cancels P

Take $\omega = e^{2\pi i/3}$ with $A + \omega B + \omega^2 C = 0$ (CCW, oly-01). Put $u = P - A$, $v = P - B$, $w = P - C$; then

$$u + \omega v + \omega^2 w = P(1 + \omega + \omega^2) - (A + \omega B + \omega^2 C)$$

$$0$$

Rotate this one identity to read all three triangle inequalities: as it stands, $u = -\omega v - \omega^2 w \Rightarrow |u| \leq |v| + |w|$; multiplied by ω^2: $|v| \leq |u| + |w|$; by ω: $|w| \leq |u| + |v|$. So PA, PB, PC form a triangle. Equality in $|u| \leq |v| + |w|$ needs $\omega v, \omega^2 w$ parallel with the same sign — precisely P on the arc BC of the circumcircle; by symmetry, on the whole circle. ∎

Concrete: equilateral $1, \omega, \omega^2$ (circumcircle $= |z| = 1$): every $P = e^{i\theta}$ makes $PA + PB = PC$ for the largest distance (degenerate), while $|P| = 2$ gives a genuine triangle (e.g. at $\theta = \frac{\pi}{3}$: $1.414 + 2.504 > 3.120$ ✓).

STANDARD APPLICATION

Why these theorems are the same trick

Gauss–Lucas: expand $\frac{p'}{p}$ and take real parts. Pompeiu: expand the ω-relation and take moduli. Ptolemy (oly-02): expand an identity and take moduli. ***Expand, then extract a geometric statement from an algebraic equality/inequality*** — the complex-bash method in three costumes. When an olympiad problem smells like "points, distances, and a hidden regular polygon", this is the family to try first.

COMMON TRAP

Traps — structure theorems

- Gauss–Lucas bounds by the convex **hull**, not the circumcircle ("zeros of P' lie in the circle through the roots" is a weaker, half-remembered statement that is sometimes false).
- The hull can be a segment (collinear roots) or the full triangle; boundary cases (critical point *on* the hull edge) are allowed by the theorem.
- Pompeiu degeneracy is *exactly* on the circumcircle — inside the triangle the distance-triangle is strictly non-degenerate (so "PA, PB, PC can never form a triangle" is wrong for interior points).
- Pompeiu needs **equilateral**; for general triangles the three distances need not satisfy any clean inequality beyond the obvious pairwise ones.
- The orientation in $A + \omega B + \omega^2 C = 0$ is CCW; swapping B, C swaps $\omega \leftrightarrow \omega^2$ and the Pompeiu rotation bookkeeping follows suit.